

Clinical Question 1: For which acne forms is systemic isotretinoin indicated?

Summary of Acne Severity and Isotretinoin Indications

Acne Type	Clinical Features	Predominant Lesion Type	Severity
----- ----- ----- -----			Comedonal
Predominantly comedones (open and closed)	Comedonal	Mild to moderate	
Mild to moderate papulopustular	Papules and pustules, few comedones		
Inflammatory or mixed	Mild to moderate	Severe papulopustular (may have some nodules)	
Widespread papules, pustules, comedones, occasional nodules			
Inflammatory/mixed	Severe	Very severe (nodular/conglobate)	Extensive inflammation, nodules, cysts, scarring
Nodular/conglobate	Very severe		

Grading: - **4: Severe** – Entire face involved with numerous papules, pustules, comedones, and rare nodules. - **5: Very Severe** – Highly inflammatory acne with nodules and cysts; may involve face, trunk, or both.

Indications for Systemic Isotretinoin

European Directive (2016 S3 Guidelines)

- **Recommended for:** - Severe papulopustular acne - Moderate to severe nodular or conglobate acne
- **Strength of Recommendation (SOR):** High
- **Positioning:** Second-line therapy after failure of antibiotics and topical treatments
- **Important Note:** Oral isotretinoin should only be used in severe, nodular, or conglobate acne that is unresponsive to appropriate topical and systemic therapies.
- **Legal Consideration:** Deviating from this guidance may expose prescribers to liability if adverse outcomes occur.

Canadian Clinical Practice Guideline (2016)

- **Recommended for:** Severe acne (including cystic/conglobate)
- **Strength of Recommendation:** High
- **Positioning:** First-line therapy in severe, scarring, or refractory acne
- **Rationale:** Based on high efficacy, long-term remission rates, safety profile, and patient preference
- **Definition of High SOR:** Action strongly recommended; definitely use

Expert Consensus and Flexibility

- Despite the European directive, many experts support **first-line use** of isotretinoin in:
 - Severe, scarring acne
 - Acne with significant psychosocial impact
 - Acne unresponsive to conventional therapy (refractory)
 - Patients at high risk for scarring
- **Limitations** (e.g., cost, access, legal or licensing issues) may require use of lower-SOR treatments even when isotretinoin is clinically preferred.

Dosing Recommendations

Parameter	European Guidelines	Canadian Guidelines	Expert Consensus
			Daily Dose – 0.5–1.0 mg/kg/day
	Start ≤ 0.5 mg/kg/day; increase to 1 mg/kg/day as tolerated		
	Often: 0.5 mg/kg for first 4 weeks, then 1 mg/kg		Dosing Frequency
Not specified (once vs. twice daily)	Not specified	Splitting dose may improve tolerance	
	Cumulative Dose Target – 120–150 mg/kg (associated with lower relapse rates)		
	Duration of Treatment 16–24 weeks for nodular/conglobate acne	Until full clearance + 1 month	Individualized; typically 16–20 weeks

Key Takeaways

- **Isotretinoin** is indicated for severe, nodular, conglobate, or scarring acne.
- **European guidelines** position it as **second-line** after antibiotic failure.

- **Canadian and expert consensus guidelines support first-line use in severe or high-risk cases.**
- **Dosing:** Start low (≤ 0.5 mg/kg), escalate to 0.5–1 mg/kg/day; aim for cumulative dose of 120–150 mg/kg.
- **Treatment duration:** Continue until complete clearance plus ~ 1 month, or for at least 16–24 weeks.
- **Clinical judgment, patient factors, and access limitations** influence real-world prescribing.